

Component-Based Software Engineering (CBSE)

Portfolio Section: CBSE Strategy

1. Positioning Your System in CBSE

Approach: Development with Reuse

Justification: For the Najm application, we focus on integrating existing, high-quality components rather than building every functionality from scratch. This decision is driven by several planning factors:

- **Market Availability:** Highly mature components for Geospatial tracking (Google Maps API) and Identity Verification (Nafath) are readily available.
- **Development Schedule:** Using off-the-shelf components allows for rapid deployment of complex features like instant damage assessment, meeting tight deadlines.
- **Budget:** It is more cost-effective to license specialized, secure services than to develop and maintain them in-house.

2. Component Composition & Interface Design

Independent Components:

- Identity Component (Nafath): Manages secure user authentication.
- Mapping & Location Component (Google Maps SDK): Handles geofencing and incident location.
- Insurance Verification Component (ELM): Validates vehicle and policy details.

Interface Design (Requires/Provides):

To ensure seamless integration, the components communicate via structured interfaces:

- The Claims Engine (Core): Requires "getUserIdentity()" from Nafath and "validatePolicy()" from ELM. It Provides "reportAccidentData()" to the insurance provider.
- The Identity Component: Provides "authenticateUser()" to the Claims Engine.
- The Mapping Component: Provides "getCoordinates()" and "mapView()" to the user interface.

3. Validation & Risk Measurement

Integration Testing Plan: We will utilize "Interface Testing" and "Smoke Testing" in a sandbox environment. This ensures that the system handles both successful responses and error codes (like API timeouts) from external components without system-wide failure.

Potential Risk: API Version Mismatch / Validation Failure.

Mitigation Strategy: We will implement the "Adapter Design Pattern." This creates a translation layer between our core logic and external APIs, allowing us to update only the adapter if a component provider changes their interface format.

4. Industry Alignment: Composable AI

Our strategy leverages "Composable AI." By integrating off-the-shelf AI components for image recognition and damage assessment, we can swap or upgrade AI models as technology advances without rewriting the central business logic, significantly improving system efficiency.

5. From Technical Knowledge to Professional Wisdom

Efficiency is not defined by the volume of code written, but by the orchestration of existing excellence. By integrating off-the-shelf components, Najm achieves a level of reliability and speed that a custom-built, untested system could not provide. This strategy ensures a resilient software ecosystem.